

Designing A Cost-Effective and Versatile Home Area Network Device

The device proposed here integrates seamlessly with a home area network and keeps a tab on the energy consumption and operation of electrical equipment while acting as a low-cost energy meter

**MANISH SHARMA, MANISH JINDGAR AND
BISWAPRAKASH NAVAJEEVAN**

The sky-rocketing cost of energy production has necessitated a more efficient energy consumption process. This has brought revolution in electrical equipment manufacturing and energy metering infrastructure. Home area network (HAN) is an advanced electrical ecosystem in which a smart utility meter and HAN devices communicate with each other to control the energy consumption profile. Armed with the

or hibernate the equipment when unused.

This low-cost device can be hooked in the existing electrical infrastructure without the need to replace, renovate, alter or rework the infrastructure. The power consumption is very low (in microamperes) when it's not in use. The following sections describe certain enhancements to the basic HAN device architecture which extend its capability and feature-set.

Architecture

The HAN device can be considered as an intelligent power socket, which at one end connects to the normal power socket and on the other end offers pluggable connection interface for home appliances, e.g., microwave and air-conditioner. It can be controlled directly by the utility meter over wireless interfaces like radio frequency

diagram of the HAN device. The device consists of a microcontroller, 230V-3.3V converter, relay, signal conditioning circuitry, infrared (IR) interface (supporting both transmitter and receiver), LCD panel and RF/power-line communication physical layer.

Its main operational features are:

1. Very low current consumption (10 μ A) when it's not functional
2. Very low run current (10 mA); 40 mA at full load
3. Fully controlled by the energy meter
4. High-voltage cut-off to save the appliances
5. Wireless communication over 2.4GHz Zigbee
6. Month-wise information storage of the power consumed
7. Fully-functional system starting from 90V AC to 300V AC
8. Easy to hook on to the network
9. Compact in size

Role of various components

Fig. 2 shows the block diagram of the device components. The role of these components is described below:

Microcontroller. The microcontroller or system-on-a-chip (SoC) plays a pivotal role in the device operation. In addition to controlling other components, it stores the application firmware in its internal Flash memory. For supporting various functionalities of the HAN device, the microcontroller should be equipped with the following features:

1. Low-power processing core with the capability to perform complex

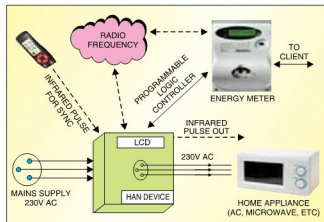


Fig. 1: Application diagram of HAN device

latest technological advancements in the field of energy utilisation, HANs are ready to supplant the traditional electrical ecosystems at home.

A basic HAN device has a two-way communication link with a utility meter and optionally with other devices in a HAN ecosystem, sharing energy consumption data of the equipment it is connected to and also receiving commands to turn off

or hibernate the equipment when unused. Additionally, its firmware can be upgraded over the RF/programmable logic controller (PLC) interface by the utility meter. Various energy parameters of the device can be displayed on the LCD. It also supports battery backup option for maintaining the time and date.

Fig. 1 shows the application